Name: Toppiiz ID: _____ No: _____

2110201 Quiz 1: Basic Skills

1. (4 points) Find the reduced row echelon of the following matrices

a. $\begin{bmatrix} 2 & 3 & 2 & 0 \\ 5 & 2 & 5 & 0 \\ 1 & 5 & 6 & 2 \end{bmatrix} \xrightarrow[-\frac{1}{2}P_1+P_3]{\frac{1}{2}P_1,\ -5/2P_1+P_2} \begin{bmatrix} 1 & 3/2 & 1 & 0 \\ 0 & -11/2 & 0 & 0 \\ 0 & 4/2 & 5 & 2 \end{bmatrix}$

$\xrightarrow[7/11P_2+P_3]{3/11P_2+P_1,\ -2/11P_2} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 2 \end{bmatrix}$

$\xrightarrow[\frac{1}{5}P_3]{-\frac{1}{5}P_3+P_1} \begin{bmatrix} 1 & 0 & 0 & -2/5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2/5 \end{bmatrix}$ #

b. $\begin{bmatrix} 2 & 4 & 2 & 2 \\ 0 & 6 & 0 & 5 \\ 3 & 2 & 3 & 3 \end{bmatrix} \xrightarrow[-\frac{3}{2}P_1+P_3]{\frac{1}{2}P_1} \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 6 & 0 & 5 \\ 0 & -4 & 0 & 0 \end{bmatrix}$

$\xrightarrow[\frac{3}{2}P_3+P_2,\ -\frac{1}{4}P_3]{\frac{1}{2}P_3+P_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

$\xrightarrow[\frac{1}{5}P_2,\ P_2 \leftrightarrow P_3]{-\frac{1}{5}P_2+P_1} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ #

2. (1 point) Write the system of linear equations defined by the following augmented matrix.

$$\begin{array}{c} \left[\begin{array}{ccccc|c} 3 & 2 & 2 & 0 & 2 & 1 \\ 1 & 1 & 2 & 3 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 3 \end{array}\right] \\ \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \end{array}$$

$$\begin{aligned} 3x_1 + 2x_2 + 2x_3 \qquad + 2x_5 &= 1 \\ x_1 + x_2 + 2x_3 + 3x_4 + 2x_5 &= 2 \\ x_2 \qquad + 2x_4 + x_5 &= 3 \end{aligned}$$

#

3. (1 point) Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 5 & 4 & 6 \end{bmatrix}$ and $x = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$.

a. Show how Ax can be computed using dot product.

b. Show how Ax can be computed as a linear combination of the columns of A.

c. Find Ax.

a) $A\vec{x} = \begin{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \\ \begin{bmatrix} 5 \\ 4 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \end{bmatrix}$ #

b) $A\vec{x} = 2\begin{bmatrix} 1 \\ 5 \end{bmatrix} + 1\begin{bmatrix} 3 \\ 4 \end{bmatrix} + 3\begin{bmatrix} 2 \\ 6 \end{bmatrix}$ #

c) $A\vec{x} = \begin{bmatrix} 11 \\ 32 \end{bmatrix}$ #

4. (4 points) Find the solution set of the systems of linear equations defined by the following augmented matrices. Compute the augmented matrices in RRE form. Identify homogeneous and particular parts of the solution. State which are pivot and free variables.

a. $\left[\begin{array}{ccccc|c} 1 & 3 & 2 & 0 & 4 & 1 \\ 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 3 \end{array}\right]$

RRE: $\xrightarrow{-3P_2+P_1} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & -2 & -5 \\ 0 & 1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 3 \end{array}\right]$

$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array}$

Pivot: x_1, x_2, x_4

Free: x_3, x_5

$$\text{Sol}^{\underline{n}}\text{ set} = \left\{ \underbrace{\begin{bmatrix} -5 \\ 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{\text{Particular Part}} + \underbrace{\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 2 \\ -2 \\ 0 \\ -1 \\ 1 \end{bmatrix} x_5}_{\text{Homogeneous Part}} \;\middle|\; x_3, x_5 \in \mathbb{R} \right\}$$

b. $\left[\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & 4 & 8 \end{array}\right]$

RRE: $\xrightarrow{\frac{1}{2}P_1} \left[\begin{array}{ccccc|c} 0 & 0 & 0 & 1 & 2 & 4 \end{array}\right]$

$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array}$

Pivot: x_4

Free: x_1, x_2, x_3, x_5

$$\text{Sol}^{\underline{n}}\text{ set} = \left\{ \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 4 \\ 0 \end{bmatrix}}_{\text{Particular Part}} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} x_5}_{\text{Homogeneous Part}} \;\middle|\; x_1, x_2, x_3, x_5 \in \mathbb{R} \right\}$$

(Already in RRE)

c. $\left[\begin{array}{cccccc|c} 0 & 1 & 0 & 3 & 0 & 0 & 5 \\ 0 & 0 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right]$

RRE: $\left[\begin{array}{cccccc|c} 0 & 1 & 0 & 3 & 0 & 0 & 5 \\ 0 & 0 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right]$

$\begin{array}{cccccc} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{array}$

Pivot: x_2, x_3, x_5, x_6

Free: x_1, x_4

$$\text{Sol}^{\underline{n}}\text{ set} = \left\{ \underbrace{\begin{bmatrix} 0 \\ 5 \\ 5 \\ 0 \\ 6 \\ 7 \end{bmatrix}}_{\text{Particular Part}} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 0 \\ -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_4}_{\text{Homogeneous Part}} \;\middle|\; x_1, x_4 \in \mathbb{R} \right\}$$

d. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{array}\right]$

RRE: $\xrightarrow{-P_2+P_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 \end{array}\right]$

$\begin{array}{ccc} x_1 & x_2 & x_3 \end{array}$

Pivot: x_1, x_2

Free: x_3

$$\text{Sol}^{\underline{n}}\text{ set} = \phi$$